

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	18 July 2026
Team ID	XXXXXX
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

The objective of this brainstorming session is to identify innovative AI-based solutions that can automatically evaluate a learner's conceptual understanding through voice explanations instead of relying solely on traditional written examinations. The selected solution should improve learning outcomes, reduce manual evaluation effort, and provide instant personalized feedback.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Our team discussed various challenges faced in educational assessment systems. After reviewing current online learning platforms and traditional evaluation methods, we identified that most systems focus on written examinations and do not effectively assess a student's conceptual understanding through verbal communication.

Identified Problems

- Manual viva examinations consume significant faculty time.
- Students receive delayed feedback.
- Existing online platforms mainly evaluate written answers.
- No automatic system evaluates conceptual understanding through speech.
- Communication skills and confidence are rarely assessed.
- Personalized learning suggestions are generally unavailable.

Final Problem Statement

Design an AI-powered application that automatically evaluates a learner's conceptual understanding by analyzing spoken explanations, comparing them with reference content, and providing detailed feedback, reports, and improvement suggestions.

STEP-1: TEAM GATHERING, COLLABORATION AND SELECT THE PROBLEM STATEMENT



Voice-Based Concept Understanding Analyser

An AI-powered application that evaluates a learner's conceptual understanding through voice explanations and provides intelligent feedback and insights.

Brainstorm & Idea Prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

- 90 minutes to prepare
- 1 hour to collaborate
- 3-8 people recommended

1 Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

10 minutes



Team gathering

Define who should participate in the session and send an invite. Share relevant information or pre-work ahead.



Set the goal

This helps the audience Valid for focusing on solving in the brainstorming session.



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2 Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

5 minutes

PROBLEM STATEMENT

How might we automatically evaluate a learner's conceptual understanding through voice explanations and provide instant, accurate feedback and insights?



Key rules of brainstorming

- Stay in topic.
- Encourage wild ideas.
- Defer judgment.
- Listen to others.
- Go for volume.
- If possible, be visual.

Brainstorming Rules

- ✓ Encourage creative thinking
- ✓ Focus on educational impact
- ✓ Prioritize AI-driven solutions
- ✓ Consider technical feasibility
- ✓ Ensure user-friendly design
- ✓ Support future scalability

Step-2: Brainstorm, Idea Listing and Grouping

Individual Ideas

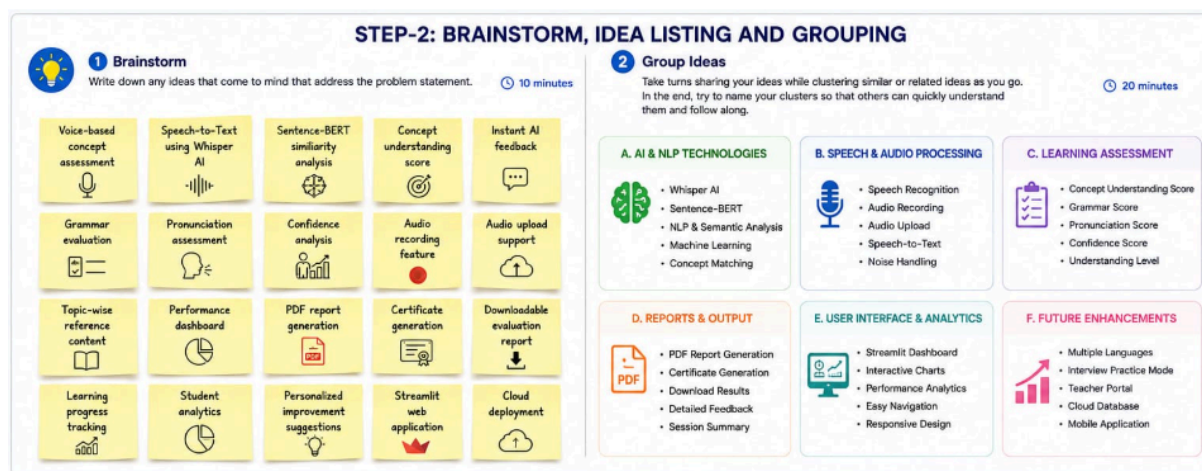
Voice-based concept assessment
AI-powered speech evaluation

Automatic viva examination
Whisper Speech-to-Text
Audio recording
Audio upload support
Sentence-BERT similarity comparison
Semantic analysis
NLP-based concept matching
Grammar evaluation
Pronunciation analysis
Confidence assessment
AI-generated personalized feedback
Performance dashboard
Student progress tracking
PDF report generation
Certificate generation
Downloadable evaluation reports
Topic-wise reference content
Multiple subject support
Teacher dashboard
Streamlit web application
Responsive interface
Cloud deployment

Group Ideas

Artificial Intelligence
Whisper AI
Sentence-BERT
NLP
Machine Learning
Speech Processing
Speech Recognition
Voice Recording
Audio Upload
Speech-to-Text
Learning Assessment
Concept Understanding Score
Semantic Similarity
Grammar Score
Understanding Level
Reports
PDF Report
Certificate
Download Results
User Interface

Streamlit Dashboard
Interactive Charts
Performance Analytics
Easy Navigation
Future Enhancements
Multiple Languages
Interview Practice
Teacher Portal
Cloud Database
Mobile Application



Step-3: Idea Prioritization

Technologies Selected

Programming Language

Python

Framework

Streamlit

Speech Recognition

OpenAI Whisper

Natural Language Processing

Sentence-BERT

AI Techniques

Semantic Similarity

Speech-to-Text

Text Analysis

Concept Evaluation

Report Generation

ReportLab PDF

